

JUN YOUNG KIM

Ph.D. Student, North Carolina State University

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EDUCATION

North Carolina State University

Ph.D., Department of Mechanical and Aerospace Engineering

Aug. 2025 – present

NC, United States

Hanyang University

Master of Science, Department of Interdisciplinary Robot Engineering Systems

Mar. 2014 – Aug. 2022

Bachelor of Science, Department of Robotics

Ansan, Republic of Korea

RESEARCH INTERESTS

Humanoid Loco-Manipulation
Safety-Critical Control

Hybrid Learning and Control
Contact-rich manipulation

Whole-Body Control
Optimization and RL

PUBLICATIONS

Manuscripts and Preprints

[m2] **J. Y. Kim**, J. Lee. “An Integrated Framework for Safe and Balanced Humanoid Manipulation via Contact-Aware Planning and Whole-Body Control,” manuscript submitted to *IEEE Robotics and Automation Letters (RA-L)*, 2026.

[m1] **J. Y. Kim**, J. An, Y. Oh. “Reactive Locomotion Control for a Wheeled-Legged Humanoid Robot Leveraging Task-Prioritized Whole-Body Control,” 2025

Journal Articles

[j1] J. An, **J. Y. Kim**, M. T. Lim, Y. Oh. “High-Speed and Enhanced Motion Control for a Wheeled-Legged Humanoid Robot Using a Two-Wheeled Inverted Pendulum With Roll Joint,” *IEEE Access*, 2025 [doi]

Conference Papers

[c4] S. H. Lee, D. Noh, **J. Y. Kim**, J. Yong, J. Y. Kim, J. K. Han. “Enhanced Localization for Bipedal Humanoid Robots in Dynamic and Noisy Environments,” *2025 10th International Conference on Control and Robotics Engineering (ICCRE)*, 2025 [doi]

[c3] **J. Y. Kim**, M. S. Ahn, J. K. Han. “Enhancing AdultSize Humanoid Localization Accuracy: A Vision-based aMCL Leveraging Object Detection Model and Hungarian Algorithm,” *2023 IEEE-RAS 22nd International Conference on Humanoid Robots (Humanoids)*, 2023 [doi]

[c2] H. J. Kim, D. K. Oh, **J. Y. Kim**, M. S. Kang. “Development of a Sensor Module for Proximity Communication and Obstacle Detection of Handrail Mounted Type Walking Assists,” *Korean Society for Precision Engineering*, 2018

[c1] H. J. Kim, H. J. Kim, D. K. Oh, **J. Y. Kim**, H. J. Byun, M. S. Kang. “Learning to Generate Trajectory of Striking Motion for Two-Link Robot Arm in Air Hockey,” *Korean Society for Precision Engineering*, 2017

RESEARCH EXPERIENCE

Graduate Researcher | Hybrid Intelligent Experimental Robotics (HIER) Lab

North Carolina State University (Advisor: Prof. Jaemin Lee)

Aug. 2025 – present

NC, United States

- **Hybrid Learning and Model-Based Loco-Manipulation:** Developing humanoid loco-manipulation methods that combine learning-based policies with model-based planning and whole-body control.
- **Kinodynamic Contact Planning:** Proposed an integrated framework that combines feasibility analysis, contact planning, and whole-body control for safe and balanced task execution. [project]

Research Associate | Humanoid Robotics Laboratory

Mar. 2023 – Dec. 2024

*Korea Institute of Science and Technology (Advisor: Dr. Yonghwan Oh)**Seoul, Republic of Korea*

- **Dynamic Locomotion for Wheeled-Legged Humanoid:** Developed a DCM-based online footstep planner that reactively computes foot placement and step timing. [[demo](#)]
- **Hybrid Locomotion for Wheeled-Legged Humanoid:** Unified walking and driving modes using convex MPC to improve high-mobility locomotion. [[demo](#)]

Visiting Graduate Researcher | Robotics & Mechanisms Laboratory

Jan. 2022 – Jul. 2022

*University of California, Los Angeles (Advisor: Prof. Dennis Hong)**CA, United States*

- **Humanoid Localization in Soccer Field:** Developed a vision-based localization system for a humanoid robot platform by integrating a particle filter with an object-detection model. [[demo](#)]

Graduate Researcher | HERoEHS Laboratory

Mar. 2020 – Jan. 2022

*Hanyang University (Advisor: Prof. JeaKweon Han)**Ansan, Republic of Korea*

- **Adult-Size Humanoid Robot Development:** Developed a kinematic controller for humanoid robot motion generation and a computer vision-only localization method.
- **Robot Systems Integration:** Integrated UWB sensors into balloon-robot systems for multi-robot positioning and coordinated field deployment.
- **Safer Robot Interaction:** Led software and hardware teams to develop a mecanum wheeled robot; designed loadcell-based force response control with a Mass-Spring-Damper model. [[demo](#)]
- **Autonomous System for Dual-Arm Mobile Manipulator:** Implemented solid-state LiDAR with LOAM-based algorithm for indoor 3D SLAM.

AWARDS

NC State University Graduate Fellowship (Office of the Provost)	2025–2026
RoboCup 2022 Bangkok Humanoid League AdultSize Soccer Competition, 2nd Place	2022
RoboCup Korea Open 2020 Humanoid AdultSize Soccer, 1st Place	2020

TEACHING AND MENTORING

Teaching Assistant , MAE 405: Controls Lab, North Carolina State University	Aug. 2025 – present
Teaching Assistant , Innovation Center for Engineering Education	2020 - 2021
Student Mentor , Gangwon Career Education Institute	Jun. 2019 - Aug. 2019
Education Mentor , Hanyang ERICA Summer School 2019 (HESS2019)	Jul. 2019

CERTIFICATIONS

Nominated, Smooth Sailing Cohort, NC State College of Engineering	2025–2026
H-Mobility Class (Autonomous Driving Course), Hyundai Motor Company	2022
Accreditation Board for Engineering Education of Korea (ABEEK) Completion	2020